

Concurso de Projetos de I&D em Todos os Domínios Científicos - 2022	Estado	Fecha a
2022.04180.PTDC	A Conceder	11.03.2022 17h00 GMT

Dados gerais

Investigador responsável

Tânia Rodrigues Pereira Ramos

Título do projeto em português

WSmart Route+ : Sistema inteligente de planeamento de rotas de recolha de resíduos

Título do projeto em inglês

WSmart Route+ : Towards a Smart Waste Collection Route Planning System

Acrónimo do projeto

WSmart Route +

Palavras-chave em português

- Gestão de resíduos
- Roteamento dinâmico
- Otimização
- Sensores volumétricos

Palavras-chave em inglês

- Waste Management
- Dynamic Routing
- Optimization
- Volumetric Sensors

Tipologia do projeto

Projetos de Investigação Científica e Desenvolvimento Tecnológico

Área científica principal

Domínio científico

Ciências da Engenharia e Tecnologias

Área científica

Engenharia Mecânica

Subárea científica

Engenharia de Sistemas

Painel de avaliação

Mechanical Engineering and Engineering Systems

Fundamentação da seleção da área científica

The WSmart Route+ project studies a new paradigm on smart waste management where it explores an **Engineering Systems** approach, that combines data analytics methods with Operational Research techniques to build a tool that can support decision-making in waste management systems. This tool aims to improve waste collection operations (by increasing resources efficiency and increasing the service level provided to the citizens) supporting the adoption of a smart waste collection based on dynamic routes as opposed to the traditional blind collection based on static routes. This project focus, as well as methodologies fall into to the **Engineering Systems** scientific area since it combines engineering with perspectives from industrial management in order to address the design and development of complex, large-scale real systems.

Calendarização

Data de início

02.01.2023

Duração (meses)

36

Instituição proponente

Instituição	Unidade de I&D
Associação do Instituto Superior Técnico para a Investigação e o Desenvolvimento	→ Associação do Instituto Superior Técnico para a Investigação e o Desenvolvimento - Centro de Estudos de Gestão do IST

Descrição da instituição e respetivas competências para o desenvolvimento do projeto

The **Associação do Instituto Superior Técnico para a Investigação e Desenvolvimento** (ISTID) is a private nonprofit organization dedicated to the development of science and technology-related activities, fostering knowledge transfer and promoting the involvement of national and foreign researchers in RD&I activities and projects. ISTID promotes RD&I activities in major areas of expertise related to challenges with a strong societal impact being strongly interdisciplinary, cutting across various fields of engineering, science & technology and architecture. The main areas of intervention are Basic Sciences, Information Technologies and Communications, Energy, Environment and Mobility, Applied Life Sciences, Materials, Nanotechnologies and Nanosciences, Technology Management, Entrepreneurship and Engineering and Manufacturing Technologies.

CEGIST, the Center for Management Studies of Instituto Superior Técnico (IST) is a multidisciplinary research center hosted by ISTID. CEGIST explores and reinforces the links between management science and engineering approaches to stimulate creativity in problem-solving, challenged by sustainability paradigms through its 5 research lines of research: DECISion science and management engineerING, Systems modelling and Methods of Operations reSearch And analytICs, Strategy, Entrepreneurship and Innovation, Operations, Logistics and Supply Chain Management, and Strategic approaches to ENvironment and SUsustainability.

Instituições participantes

País	Instituição	Unidade de I&D
PORTUGAL	Universidade de Lisboa (UL) - Instituto Superior Técnico - Instituto de Engenharia de Sistemas e Computadores - Instituto de Engenharia de Sistemas e Computadores, Investigação e Desenvolvimento em Lisboa	→ Universidade de Lisboa (UL) - Instituto Superior Técnico - Instituto de Engenharia de Sistemas e Computadores - Instituto de Engenharia de Sistemas e Computadores, Investigação e Desenvolvimento em Lisboa

Descrição da instituição e respetivas competências para o desenvolvimento do projeto

The Instituto de Engenharia de Sistemas e Computadores, Investigação e Desenvolvimento em Lisboa (INESC-ID) is a R&D institute dedicated to advanced research and development in the fields of Information Technologies, Electronics, Communications, and Energy.

INESC-ID is a non-profit institution, privately owned by IST and INESC, officially declared of public interest. It was created in 2000, as a result of a reorganization of its parent institution. Since December 2004, the institution has the status of “Laboratório Associado” from FCT.

Currently, we integrate more than one hundred PhD researchers and two hundred post-graduation students divided between nineteen research groups, organized in five main research lines: Computing Systems and Communication Networks; Embedded Electronic Systems; Information and Decision Support Systems; Interactive Intelligent Systems; Energy Systems.

INESC-ID has participated in more than 50 research projects funded by the European Union and more than 190 funded by national entities. Until today, our researchers have published more than 700 papers in international journal papers, more than 3000 papers in international conferences, and have registered 15 patents and/or brands.

País	Instituição	Unidade de I&D
PORTUGAL	Universidade de Coimbra	→ Universidade do Porto - Faculdade de Engenharia da Universidade do Porto - Centro de Investigação do Território, Transportes e Ambiente

Descrição da instituição e respetivas competências para o desenvolvimento do projeto

The University of Coimbra (UC) was founded in 1290 and constitutes a reference in higher education and scientific research in Portugal, with excellent academic reputation recognized worldwide. It includes 8 faculties and over 30 research units and centres, providing teaching and conducting research in almost all study fields - Exact Sciences and Engineering, Life Sciences, Architecture, Medicine, Social Sciences, Humanities and Sports -, as well as several structures devoted to promote science and technology, culture and sports.

UC pursues a policy of continuous improvement in several areas, ensuring high standards of teaching and research, as well as being an active participant in the development of the corporate world. The continuous technological development promoted by UC’S different R&D units, addresses the challenges involved in the design, operation and regulation of technologies, bringing together faculty from different scientific disciplines with a long experience in teaching, research, technology transfer and consultancy.

UC stands out also on technology transfer being involved in the promotion of several initiatives, like IPN Incubator and Biocant, and in the creation of highly successful technological spin-offs, as Active Space Technologies, Crioestaminal, Critical Software, Feedzai, amongst many others.

País	Instituição	Unidade de I&D
PORTUGAL	EVOX Technologies	→ EVOX Technologies

% de autofinanciamento no orçamento

50

Descrição da instituição e respetivas competências para o desenvolvimento do projeto

EVOX Technologies is a visionary and devoted team that develops high quality and world class mechanisms for today’s industry. The mission is to investigate and develop cutting edge products for the industrial markets. Our products and services offer competitive advantages over the competition taking in consideration innovation and excellence. Our team is composed by 12 employees in which 5 are MScs in the areas of Computer Science, Electronics and Telecommunications, committed in carry out world class research and innovation resulting in future integration in EVOX Technologies products.

As a knowledge eager SME, EVOX Technologies core competences cover a wide array of expertise, which enables us to better address customer requirements and engineering solutions for each project profile. EVOX Technologies is independent from large manufacturers and proprietary platform providers. Being vendor independent, enables us to accurately select and recommend the most suitable and cost-effective third-party technologies to each customer specific requirements. Also, our team works in close partnership with national and international R&D organizations and local academic institutions.

Instituições de colaboração

-



Investigador responsável

Tânia Rodrigues Pereira Ramos Nuclear IR 40% de dedicação



Ciência ID
9C1F-3BB3-36FB

Instituição à qual está associado no âmbito do projeto de investigação
Associação do Instituto Superior Técnico para a Investigação e o Desenvolvimento

CV
Obtido em 06-03-2022 at 18:40:54 GMT

Sinopse do CV
Tânia Ramos is an Associate Professor at the Engineering and Management Department at IST – University of Lisbon, and Vice-President of the Research Unit CEGIST. She holds a BSc degree in Management (ISCTE-IUL), a MSc degree in Operations Research and Systems Engineering (IST) and a PhD in Engineering and Management (IST). Tânia’s research focuses mainly on modelling and solving real-world problems related to logistics systems planning, routing, sustainability, reverse logistics and waste management systems. Tânia is co-author of 1 book, 8 book chapters (6 international books), and 12 peer-review articles published in indexed international journals all in Q1 from their areas of research (4 Q1: Operations Research & Management Science, 4 Q1: Engineering Environmental and Environmental Sciences according with Clarivate Analytics Web of Science and 4 Q1: Management Science and Operations Research according with SCImago Journal Ranking).

The impact of Tânia’s research is acknowledged by the increasing number of citations over the years (487 citations since 2012 - the year she finished her PhD, and 107 citations in 2021), with an h-index of 9 (Scopus). Two of her most cited works are the papers entitled “Research challenges in municipal solid waste logistics management” published in Waste Management Journal in 2016 (IF 7.145), with 137 citations, and “Planning a sustainable reverse logistics system: Balancing costs with environmental and social concerns” published in Omega Journal in 2014 (IF 7.084), with 139 citations.

Tânia coordinated several national projects on the fields of logistics and waste management, namely she was the PI of the FCT-Funded “WSMART Route - Waste Collection Real-Time Route Planning System” (MIT-EXPL/SUS/0132/2017) and of the ERSUC-funded project “Logistics Optimization of Selective Waste Collection using Transfer Stations”, and the Co-PI of the Open Innovation Call by Sociedade Ponto Verde-funded project “Mobile-Pro-U: Integrated system based on a mobile processing unit to recover packaging glass from undifferentiated urban waste”.

Most of her papers are related to this project. The most recent ones and specifically related to the smart waste management topic are:

- A hybrid metaheuristic for smart waste collection problems with workload concerns (2022) 10.1016/j.cor.2021.105518
- Searching for a Solution Method for the Smart Waste Collection Routing Problem (2021) 10.1007/978-3-030-85476-8_1
- Dynamic Approaches to Solve the Smart Waste Collection Routing Problem (2019) 10.1007/978-3-030-10731-4_12
- The smart waste collection routing problem: Alternative operational management approaches (2018) 10.1016/j.eswa.2018.03.001
- Waste collection planning based on real-time information (2018) 10.1007/978-3-319-71583-4_22

Her ability to engage young researchers is also notable by the high number of thesis supervisions. Tânia supervised more than 50 Master's dissertations, 1 PhD dissertation and has 2 ongoing PhD students.

Custo total (em euros)

2023	2024	2025	2026	Total
0,00 €	0,00 €	0,00 €	0,00 €	0,00 €

Membros

Ana Paula Ferreira Dias Barbosa Pova Nuclear Membro 15% de dedicação



Associação
Confirmada

Ciência ID
651C-2DDA-A52C

Instituição à qual está associado no âmbito do projeto de investigação
Associação do Instituto Superior Técnico para a Investigação e o Desenvolvimento

CV
Obtido em 23-02-2022 at 18:35:12 GMT

Custo envolvido (em euros)

2023	2024	2025	2026	Total
0,00 €	0,00 €	0,00 €	0,00 €	0,00 €

António José Pais Antunes Nuclear Membro 20% de dedicação

Associação
Confirmada

Ciência ID
4917-DB4C-D77D

Instituição à qual está associado no âmbito do projeto de investigação
Universidade de Coimbra

CV
Obtido em 25-02-2022 at 18:14:43 GMT

Custo envolvido (em euros)

2023	2024	2025	2026	Total
0,00 €	0,00 €	0,00 €	0,00 €	0,00 €

Diana Rita Ramos Jorge Membro 10% de dedicação

Associação
Confirmada

Ciência ID
9F17-52AF-9FBD

Instituição à qual está associado no âmbito do projeto de investigação
Associação do Instituto Superior Técnico para a Investigação e o Desenvolvimento

CV
Obtido em 07-03-2022 at 15:30:31 GMT

Custo envolvido (em euros)

2023	2024	2025	2026	Total
0,00 €	0,00 €	0,00 €	0,00 €	0,00 €

Hélio Silva Membro 15% de dedicação

Associação
Confirmada

Ciência ID
9B19-F708-C33C

Instituição à qual está associado no âmbito do projeto de investigação
EVOX Technologies

Tipo de vínculo
Contrato de trabalho

CV
Obtido em 27-02-2022 at 19:35:13 GMT

Custo envolvido (em euros)				
2023	2024	2025	2026	Total
1 740,38 €	1 484,44 €	3 531,94 €	0,00 €	6 756,76 €

Manuel Lopes Membro 15% de dedicação

Associação
Confirmada

Ciência ID
0A12-41BD-718F

Instituição à qual está associado no âmbito do projeto de investigação
Universidade de Lisboa (UL) - Instituto Superior Técnico - Instituto de Engenharia de Sistemas e Computadores - Instituto de Engenharia de Sistemas e Computadores, Investigação e Desenvolvimento em Lisboa

CV
Obtido em 08-03-2022 at 12:51:00 GMT

Custo envolvido (em euros)				
2023	2024	2025	2026	Total
0,00 €	0,00 €	0,00 €	0,00 €	0,00 €

Pedro Alexandre Simões dos Santos **Nuclear** Co-investigador responsável 25% de dedicação

Associação
Confirmada

Ciência ID
4B1A-8583-BB76

Instituição à qual está associado no âmbito do projeto de investigação
Universidade de Lisboa (UL) - Instituto Superior Técnico - Instituto de Engenharia de Sistemas e Computadores - Instituto de Engenharia de Sistemas e Computadores, Investigação e Desenvolvimento em Lisboa

CV
Obtido em 27-02-2022 at 19:54:56 GMT

Custo envolvido (em euros)				
2023	2024	2025	2026	Total
0,00 €	0,00 €	0,00 €	0,00 €	0,00 €

Vitor Goulão Membro 15% de dedicação

Associação
Confirmada

Ciência ID
3B10-A1BE-73D6

Instituição à qual está associado no âmbito do projeto de investigação
EVOX Technologies

Tipo de vínculo
Contrato de trabalho

CV
Obtido em 07-03-2022 at 15:13:23 GMT

Custo envolvido (em euros)

2023	2024	2025	2026	Total
0,00 €	1 689,19 €	2 598,75 €	0,00 €	4 287,94 €

Novas contratações

(BI-a - 01) Bolsa de Investigação - Estud. dout. ou lic. e mestres em CNCG100 % de dedicação

Instituição
Associação do Instituto Superior Técnico para a Investigação e o Desenvolvimento

Tarefas
Task 2 - Smart Waste Collection Routing Problem Modelling · 21 pessoa * mês
Task 5 – Cost-Benefit Analysis · 8.25 pessoa * mês
Task 6 - Prototype Tool Validation · 6.75 pessoa * mês

Custo envolvido (em euros)

2023	2024	2025	2026	Total
16 136,00 €	16 136,00 €	16 136,00 €	0,00 €	48 408,00 €

(BI-b - 01) Bolsa de Investigação - Estud. mestrado, mestrado int. ou lic. e mestre em CNCG17 % de dedicação

Instituição
Associação do Instituto Superior Técnico para a Investigação e o Desenvolvimento

Tarefas
Task 4 - Strategic Location for Sensors Placement · 6 pessoa * mês

Custo envolvido (em euros)

2023	2024	2025	2026	Total
0,00 €	6 455,88 €	0,00 €	0,00 €	6 455,88 €

(BI-a - 02) Bolsa de Investigação - Estud. dout. ou lic. e mestres em CNCG100 % de dedicação

Instituição
Instituto de Engenharia de Sistemas e Computadores, Investigação e Desenvolvimento em Lisboa

Tarefas
Task 1 - Data Analytics and Predictive Models · 12 pessoa * mês
Task 4 - Strategic Location for Sensors Placement · 12 pessoa * mês
Task 5 – Cost-Benefit Analysis · 6 pessoa * mês
Task 6 - Prototype Tool Validation · 6 pessoa * mês

Custo envolvido (em euros)

2023	2024	2025	2026	Total
16 136,00 €	16 136,00 €	16 136,00 €	0,00 €	48 408,00 €

(BI-a - 03) Bolsa de Investigação - Estud. dout. ou lic. e mestres em CNCG83 % de dedicação

Instituição
Universidade de Coimbra

Tarefas
Task 3 - Decision Supporting Tool for the Smart Waste Collection Routing Problem · 18 pessoa * mês
Task 5 – Cost-Benefit Analysis · 5.25 pessoa * mês

Custo envolvido (em euros)

2023	2024	2025	2026	Total
7 586,00 €	15 172,00 €	15 307,00 €	0,00 €	38 065,00 €

Consultores

Sinopse do cv da equipa

The team’s project involves 7 different entities (3 research centres CEGIST, INESC-ID and CITTA, an international expert in routing problems from the U.Bologna, an industrial IT partner (EVOX) and 2 waste collection companies (ERSUC and Valorsul) composed by a team of 8 researchers (Tânia Ramos, Ana Póvoa, Diana Jorge, Pedro Santos, Manuel Lopes, António Pais Antunes, Hélio Silva, Vítor Goulão). TR, AP and DJ belong to CEGIST, PS and ML to INESC-ID, APA to CITTA, and HS e VG to EVOX. EVOX has expertise in the monitoring technology involved; INESC-ID has expertise in data science and machine learning techniques; CITTA excels in transportation systems and solution methods, whereas CEGIST outstands in dealing with operations management, mathematical models and waste routing problems. Additionally, the internationally recognized expert in routing problems (Daniele Vigo) will act as consultant. Finally, two waste collection companies are also involved, who represent the main beneficiaries of the project’s outcomes: 1) ERSUC currently with low maturity towards a smart collection operation (manual planning, static routes), and 2) Valorsul which already gave a small step towards a smart collection operation by installing some sensors in their waste bins, although the routes are still static. Please see the supporting letters attached.

The project is grounded on multiple skills and is very multidisciplinary. Partners have worked together in the past on a one-year funded exploratory project that set the stage for the project WSmart Route+.

Ana Póvoa is a Full Professor in Operations and Logistics, at the Engineering and Management Department of Instituto Superior Técnico (IST), University of Lisbon. She holds a PhD from the Imperial College of Science Technology and Medicine and her research focus is on developing a comprehensive understanding of complex problems in supply chains and operations management, supported by novel and sound engineering systems models and techniques. Sustainability, resilience, uncertainty and risk in the design and planning of supply chains are among the main research addressed domains. Ana has created and coordinates the research group in Operations Management and Logistics (OpLog) at CEGIST. Her research has led to several national and international awards having received recently the honour of being considered the best researcher in Industrial Management at the University of Lisbon.

Diana Jorge is an Assistant Professor of Industrial Engineering and Management at the U. Lusofona. Diana holds a PhD in Transportation Systems from the U. Coimbra. Her research focuses mainly on modelling and solving real-world problems related to transportation and logistics systems, e.g. waste collection routing, through Operations Research methods. Diana is co-author of 8 peer-review articles published in indexed international journals with 561 citations (122 citations and an h-index of 6 in 2021) in Scopus, and participated in 3 research projects.

Pedro Santos has a PhD in Mathematics from the Technische Universität Chemnitz, and Habilitation in Mathematics. He is Associate Professor at both the Mathematics and the Computer Science Departments of IST. His main academic interests are in Functional Analysis, Artificial Intelligence, and Game Design, having focused in the last years on Machine Learning and its applications to several domains. Those domains include decision-making, the creation of socially intelligent agents, and the Legislative domain, where he is exploring advanced Neuro-symbolic techniques. He has co-authored 3 books and has 80 publications and 190 citations registered on Publons, and more than 700 citations and h-index 15 on Google Scholar.

Manuel Lopes is an Associate Professor in Computer Science and Engineering at IST. He did his PhD in Electrical and Computer Engineering in the field of machine learning and intelligent machines. During his career he was a researcher at VTT Finland, a lecturer at Plymouth University and a researcher at Inria France. Manuel's research interests lie in the application of machine learning methods for problems of computational biology, robotics, sensing technologies, human-machine collaboration and data analysis. He published more than 50 papers in international journals and conferences for the domains of data science and machine learning.

António Pais Antunes is a Full Professor of Spatial Planning and Transport Systems at the U. Coimbra, and has been an Invited Professor at EPF Lausanne (2010/11), and a Visiting Professor at MIT, USA (2013/14). He holds a PhD in Applied Sciences from UC Louvain. Over his career he has been involved in 22 research projects (6 times as PI) and supervised 24 completed PhD thesis. A significant part of his research has focused on waste management topics, from the planning of regional systems to the location of major treatment facilities and the design of recyclables collection routes.

Hélio Silva holds a MSc in Information Technologies from ISCTE and a MSc in Software Development and Interactive Systems from IPCB. He holds a CISCO certification in wireless communications and data network security. He worked as a senior researcher at the Instituto de Telecomunicações (IT) and coordinated the IT software development team of an European research project FP7 ROMEO. In 2015 he founded the company EVOX and is responsible for EVOX's participation in 2 R&DT projects P2020 CityAction and P2020 360Waste+ and in the FP7 project FINODEX. He has extensive experience in the development of products for the waste sector and their implementation in the market.

Vítor Goulão is a Hardware engineer at EVOX since 2016. MSc in Engineering of Mobile Communications and a BSc in Electrical Engineering and Telecommunications by IPCB. He is co-author of the Patent International nº PCT/IB2013/058672 “Self tracking system and its operation method”. He has an extensive experience in hardware development and project implementation.



Resumo

Resumo em português

O projeto WSmart Route+ estuda um novo paradigma na gestão inteligente de resíduos. A análise de dados combinada com a otimização de rotas é explorada neste projeto para melhorar a operação de recolha de resíduos, apoiando a implementação de uma recolha inteligente baseada em rotas dinâmicas, em oposição à tradicional recolha cega baseada em rotas estáticas. Todos os dias toneladas de resíduos são recolhidas e, para tal, milhares de quilómetros são percorridos. Porém, existe uma elevada incerteza sobre a quantidade de resíduos nos contentores, facto que pode levar a uma operação ineficiente, na qual os recursos são usados para recolher uma pequena quantidade de resíduos. A título de exemplo, dados recolhidos junto do nosso parceiro ERSUC, uma das maiores empresas de recolha de resíduos recicláveis, mostram que nos municípios rurais 52% dos contentores de resíduos recicláveis foram recolhidos com um nível de enchimento inferior a 25% e apenas 9% com um nível de enchimento superior 75%. Além disso, as empresas de resíduos frequentemente recebem reclamações relatando transbordos ou resíduos em redor dos contentores que prejudicam a saúde pública. Este é um caso frequente em municípios urbanos, onde um grande número de transbordos é relatado. Estes dados representam um uso ineficiente de recursos nas áreas rurais e problemas de saúde pública em áreas urbanas. Esta é uma situação comum em sistemas de gestão de resíduos. Para aumentar a eficiência e simultaneamente o nível de serviço prestado aos cidadãos serão exploradas três grandes linhas de investigação.

A *primeira* foca na redução da incerteza em relação à procura (quantidade de resíduos em cada contentor). Para lidar com esse desafio, são exploradas três formas possíveis de aquisição de dados. Uma é a instalação de sensores, onde são feitas medições precisas dos níveis de enchimento, proporcionando um conhecimento a priori desses níveis. Mas como os sensores não são baratos e podem exigir um elevado investimento das empresas de resíduos, é explorado um segundo caminho que se baseia nas observações dos motoristas sobre os níveis de enchimento dos contentores antes de esvaziá-los. Esses dados podem ser usados para alimentar modelos de previsão robustos para prever os níveis de enchimento futuros. Esta abordagem tem um custo insignificante, mas é suscetível a erros humanos e são registadas poucas observações em comparação com a instalação de sensores nos contentores. Para ultrapassar o erro humano das observações dos motoristas, um terceiro caminho é explorado, onde uma nova abordagem de monitorização é proposta – o uso de sensores móveis pelos motoristas para medir com precisão os níveis de enchimento antes de esvaziar os contentores. Isso implica apenas um sensor (móvel) por motorista, mas poucas observações são registadas em comparação com o uso de sensor estático instalado nos contentores. Os modelos de previsão robustos são também necessários nesta abordagem.

No entanto, o acesso aos dados (e tratamento) não é suficiente para aumentar a eficiência das operações nem evitar transbordos. Surge, então, a *segunda* linha de investigação, em que é necessário otimizar as rotas de recolha em função desses dados. A *terceira* linha foca-se no desafio de identificar quais os contentores que devem ter um sensor estático instalado. Uma rede completa sensorizada implica um elevado investimento dado os milhares de contentores que uma rede opera, por isso este projeto explora a escolha de um número limitado de contentores para ter sensores estáticos.

Neste contexto, este projeto visa desenvolver e testar uma ferramenta para prever níveis de enchimento tendo em conta a informação dos sensores estáticos ou móveis ou dos motoristas e definir rotas ótimas de forma dinâmica, resultando em conhecimento científico inovador e benefícios práticos para a sociedade em termos de ferramenta utilizável.

O WSmart Route+ tem, então, os seguintes objetivos de investigação:

1. Estudar diferentes formas de adquirir e tratar dados sobre o nível de enchimento de forma a reduzir a incerteza associada;
2. Desenvolver e testar uma ferramenta de planeamento que combina métodos inovadores para lidar com a complexidade relacionada com a definição de rotas de recolha inteligentes, considerando dados fornecidos pelos sensores (estáticos e móveis) e pelas observações visuais dos motoristas;
3. Desenvolver modelos inovadores para decidir em quais contentores devem ser instalados sensores;
4. Desenvolver uma análise de custo-benefício comparando a situação atual com a situação futura em casos reais;
5. Desenvolver um teste-piloto em ambiente real.

Estes objetivos contribuem para aumentar a eficiência dos recursos (ODS9), reduzir o impacte ambiental da atividade de recolha de resíduos (ODS11) e diminuir o número de transbordos, melhorando a gestão do saneamento básico (ODS6). Baseado na experiência da equipa [Ram1,Mor,Jor] estima-se que a ferramenta irá reduzir os custos em 30% e aumentar o nível de serviço à população.

Resumo em inglês

The **WSmart Route+** project studies a new paradigm on smart waste management. Data analytics combined with route optimization will be explored to improve waste collection operation supporting the adoption of a smart waste collection based on dynamic routes as opposed to the traditional blind collection that is based on static routes. Everyday thousands of kilometres are travelled to collect tonnes of waste with high uncertainty on the amount of waste in each bin, a fact that can lead to inefficient collection operations where resources are used to collect often a small amount of waste. For instance, data gathered from ERSUC, one of the major recyclable waste collection companies operating in Portugal, showed that in rural municipalities 52% of the recyclable waste bins were collected with a fill level lower than 25% and only 9% with a fill level greater than 75%. On the other hand, waste companies often receive complaints reporting overflows or waste surrounding the containers which damage public health. This is often the case in urban municipalities, where a vast number of overflows are reported. Such facts represent inefficient resource usage in rural areas and public health issues in urban areas, a common situation in waste management systems. To increase efficiency and simultaneously the level of service provided to citizens three main research lines will be explored:

The *first* one focus on reducing the uncertainty regarding demand (waste inside each individual bin). To deal with this challenge, three possible ways of acquiring data are explored. One is the installation of static sensors in bins, where accurate fill levels measurements are done, providing a priori knowledge of the fill levels. But as sensors are not cheap and may require a high investment from waste companies, a second path is explored that relies on visual observations from drivers on the bin’s fill levels before emptying them. The data collected should then be used to feed robust models for predicting fill levels. This approach has a negligible cost but is susceptible to human error and few observations are recorded compared to sensor in bins. To overcome the human error from the visual observation, a third path is pursued where a new monitoring approach is proposed – the use by truck collection drivers of mobile sensors to accurately measure fill levels before emptying the bins. This only implies one (mobile) sensor per truck, but few observations are recorded compared to the use of static sensor installed in bins. Also, robust predictive models are needed in this approach.

Accessing and treating data is not however sufficient to increase operations efficiency nor avoid overflows. Then, the second research line emerges where the focus relies on optimizing collection routes accordingly with the data collected and treated.

The third research line focus on the challenge of identifying the key bins to have static sensor installed. A full sensorized network imply a high investment given the thousands of bins own by waste management systems, therefore choosing a limited number of bins to have static sensors is explored, towards a system where some bins are monitored through static sensors, some by mobile sensors, and some by visual observations.

In such context, this project aims to develop and test a tool to predict fill-levels given the information from static sensors or mobile sensors or drivers and define optimal routes in a dynamic way, delivering both innovative scientific knowledge about data-driven optimization and practical benefits to society in terms of a workable tool. The ultimate goal of this project is to increase resources efficiency and improve service levels of waste management systems contributing to a sustainable development.

The following research goals are then set to the present project:

1. Study different ways of acquiring and treat waste bin fill-level data to reduce associated uncertainty;
2. Develop and test a planning tool combining innovative methods to deal with the complexity related with the definition of smart collection routes, considering data provided by sensors (static and mobile) and by driver’s visual observations;
3. Develop innovative models to decide in which containers static sensors should be installed;
4. Develop a cost-benefit analysis comparing the as-is to the to-be in real case studies;
5. Develop a testbed in a real environment.

The project’s objectives contribute to increase resource efficiency (SDG 9), reduce environmental impact of the waste collection activity (SDG 11), and decrease the number of overflowing bins/waste surrounding the bins, improving sanitation management (SDG 6).

A multidisciplinary team was assembled that guarantees the project’s goals. Based on the team research experience [Ram1,Mor,Jor], it is estimated that the proposed planning tool will help to reduce companies’ operational costs around 30% and increase the service level to the population by decreasing the number of overflows.

Resumo para publicação

Resumo para publicação em português

-

Resumo para publicação em inglês

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Revisão da literatura

Currently, most municipalities are not utilizing smart waste management systems. Instead, they usually rely on pre-planned routes travelled at regular intervals, often designed manually, regardless of the demand [1]. This leads to routes that are far from optimal and do not adjust to demand changes [Ram1]. Studies have shown that as regions become more developed and consumer expenditure increases, so does waste generation [2,3,4]. Therefore, urban growth and national development must be accompanied by upgrades in municipal waste management systems. To address this upgrade and overcome inefficiency in waste collection, two research avenues are apparent:

1) The installation of smart sensors in bins [5,6,7] to access real-time information on bins’ fill levels that allows waste collection companies to: i) save operational costs by reducing unnecessary trips; ii) increase the quality of service to citizens, by issuing a warning when a bin is close to overflowing. Recent works have considered the use of sensor information to optimize collection routes [8,9,10], without exploring the optimization of the set of bins to be visited. In this case, a pre-definition of the set of bins to visit is performed, usually based on a minimum threshold, and only bins that reach a certain fill level are visited. We argue that a minimum fill level limit is not a good strategy to trigger a collection [Ram1] as containers with low filling levels can “be worth it” to be collected. Moreover, all these works considered a full sensorized network, which is somehow unrealistic given that sensors are not cheap and may require a considerable investment from municipalities, which are often responsible for thousands of bins.

2) The development of robust and adaptive models for predicting fill levels [1, 2, 4, 11] which depends on historical data for individual bin fill levels, which can be difficult to gather. Some companies have their driver’s recording bin fill levels every time a bin is visited. Yet, the accuracy of this information is questionable, as it has lower precision and may be prone to individual subjectivity and human error. Moreover, if fill levels are only recorded when visited, time interval between observations can be considerable, especially for recyclable waste.

We introduced the Smart Waste Collection Routing Problem (SWCRP) [Ram1]. SWCRP characterizes dynamic smart routes where each day only the **“best” bins to visit and the “best” visiting sequence is defined** by accessing real-time information on the bins’ fill-levels through sensors and a mathematical model to maximize the daily profit (the difference between the revenues related with the weight collected and the transportation costs) while considering the problem constraints (vehicle and bins capacity, etc.). The SWCRP was modeled and solved in [Ram1] as a Vehicle Routing Problem with Profits (VRPP), an approach revealed to be myopic, that highlighted the need for a larger planning period [Mor].

We then approached the SWCRP problem as an Inventory Routing Problem (IRP) [Mor], where profit was maximized for a planning horizon of 10 days, instead of for a single day (as in VRPP). Our IRP model considers waste bins’ future fill levels in the whole planning horizon. As such, average rates from historical data through simple statistics were used. By using small-size instances, we concluded that IRP models enable better results than the VRPP model, and, therefore, we will explore IRP models in this research project. This is also corroborated by [12].

As IRP integrates inventory management with routing problems over a given time horizon [13], it addresses three different decisions: when to visit a client; how much to deliver/collect; and how to combine clients in a route [14]. When considering demand information, [15] classified the IRPs into three groups: deterministic, stochastic, and dynamic. The Dynamic Stochastic IRP (DIRP) is the one that is more relevant for this research project and only two works were

identified [14] and [16]. When coupling these models with real-time information, several difficulties arise as the dynamism required to deal with updated everyday information is not straightforward. However, nowadays DIRP approaches are needed to explore access to real-time information and better support the operations’ decision-making processes. Such works are not easily found in the literature applied to waste collection having been published only two works on this subject to the best of our knowledge [17, and Mor, from our team].

We identified several gaps in the literature. Although there is a rich body of research on bin sensor systems, authors rarely contemplate alternative monitoring approaches beyond the current situation (no monitoring) and a completely sensorized network of bins. Complete sensorization is presently an unrealistic expectation due to the large investment cost and there is a lack of work to identify the optimal placement of sensors (on the bins) to minimize uncertainty. Moreover, mobile sensors or observations of the drivers were never addressed. On the other hand, there is a growing body of literature addressing predictive models for waste generation. Still, more work is needed, especially regarding day-to-day predictions on individual bins highlighting the need for i) the development of more sophisticated solutions and ii) validation of existing solutions on larger and real scales. Up to today, only a few works published explore DIRP approaches within waste collection and none of the analyzed works explored the use of predictive methods to estimate future waste bins’ fill-levels. In the published models, cost minimization is typically assumed and no concern on the profit maximization (a metric that combines the weight collected and the travelled cost) is pursued. Additionally, none of the analyzed works explored the updating of new real-time information when this becomes available. In the WSmart Route+ these gaps will be explored.

Plano de investigação e métodos

In 2020, the amount of municipal waste generated per person in the EU reached 505 kg, 4 kg more per person than in 2019 and 38 kg more than in 1995[18]. This represents tonnes of waste that need to be collected everyday through thousands of kilometers travelled and emissions of Greenhouse Gases. The high uncertainty on the waste amount in each bin can lead to inefficient collection operations where resources are used to collect often a small amount of waste. The WSmart Route+ aims to increase Waste Management Systems resources efficiency and service levels provided to the citizens contributing to a more Sustainable Development.

WSmart Route+ explores three main research streams:

1. The identification of the more adequate form of acquiring data from waste bins considering accuracy versus investment, and how to make predictions based on these data;
2. The usage of collected data as input to optimization models that will allow the definition of optimised operational approaches to collect waste while reducing costs and guaranteeing the communities service levels;
3. Which bins need to have more continuous monitoring to assure higher benefit/cost value.

To identify the more adequate form of acquiring data from waste bins considering accuracy versus investment, and how to make predictions based on these data, we will study three different monitoring waste bins methods to reduce the uncertainty associated with their fill levels: **i) Static sensors installed inside waste bins**, that allow the collection of more and accurate data but imply a high investment; **ii) visual observations from the truck collection drivers** which are less frequent, with lower accuracy, but have a negligible cost; **iii) Mobile sensors** that represent an innovative solution (to the best of our knowledge) that balances the advantages and disadvantages of the other two, giving the accuracy and precision of sensor observations at a fraction of the cost. Regardless of the source data, machine learning models will be used to properly treat data and make predictions.

But, collecting and treating data is not enough to increase efficiency and service level in waste management systems. It is necessary to plan the collection operation taken into account these data. Some IT companies are already exploring the use of static sensors coupled with tools to define routes, a fact revealing the relevance and importance of this problem. However, they typically use algorithms to first select a set of bins to visit based on the current or estimated fill-levels, and then run a Travelling Salesman Problem (TSP) or VRP algorithm to define the collection routes. Yet, several important aspects are missing in this approach, as knowing the waste amount at each bin does not necessarily mean that only full containers should be collected on a specific day, as maybe considering small additional distances may reveal more profitable situations (e.g. collect not full containers located nearby a full container); the service level provided to the population must be considered, and defining the more “profitable” route for a day does not mean maximizing profit for a time horizon (as a decision of today often has repercussions on the following days). Regarding the visual observations, an ongoing practice at waste management companies, i.e., they ask the drivers to record the fill-levels before emptying a bin (typically, using a 25%-level scale). However, that information is not currently being used by the companies to optimize the collection operation. This is something that we will explore too.

Finally, for the development of smart cities, with effective and low-cost waste management services, it is important to understand - given the large number of bins managed by waste management companies - which bins should have a more continuous monitoring to assure higher benefit/cost value. However, choosing which bins should be sensorized through static sensors to minimize uncertainty or provide the most relevant information is not straightforward and it is seldom studied in the literature. We will contribute to fill this gap.

WSmart Route+ aims to develop a smart route planning tool and assess its benefits by answering the following open questions:

1. Data: What data sources should be used to improve efficiency and service level in waste collection operation? What is the value of information from the data collected currently by the drivers? And what would be the impact of improving the accuracy of these data by replacing visual observations by mobile sensors readings? And what would be the impact of having continuous readings (from static sensors) *versus* less frequent readings (from the drivers)?
2. Planning Dynamic Routes: What are the most suitable models to deal with dynamic routes? What should be the time horizon to consider? How to deal with the uncertainty when planning for the days ahead?
3. Sensor Placement: What are the waste bins that are worth being sensorized through static sensors? What are the criteria and thresholds involved in that decision? How the collection routes must be designed for collecting the non-monitored and monitored waste bins?
4. Tool: What are the most suitable algorithms capable to deliver solutions for dynamic collection routes in a short computational time? How should the data be captured - by either drivers and sensors (static or mobile), and the solution plan presented to the decision-maker?
5. Benefits: What are the benefits of using a system with the previous features when compared to the current situation?

WSmart Route+ will integrate the experience and knowledge of the multidisciplinary research team to comprehensively answer the above open questions. For that, the following tasks were devised:

Task 1-Data Analytics and Predictive Models (month 1 to 12). To answer the [Data open question](#), we will use 2 datasets already available. The ERSUC and Valorsul drivers register fill levels of the individual waste bins before collecting. And data from static sensors readings obtained from 650 volumetric sensors already installed by Valorsul. A third dataset obtained through the mobile sensors readings will be created. For that, an experiment will be conducted in the first 6 months of the project where a set of drivers will be involved to collect the data with the fill-levels through mobile sensors. The three data sets will be then analysed through proper data analytics methods and data will be transformed into valuable information for management purposes (e.g., waste bin visits that could be avoided, overflows that could be avoided, etc.). The three data sets will also act as historical data to feed machine learning algorithms to help predicting future fill levels of individual bins. These predictive models are needed to plan dynamic routes with a “look-ahead perspective” in T2.

Tasks 2-Smart Waste Collection Routing Problem Modelling (m 1 to 21).To answer the [Planning Dynamic Routes open questions](#), we will explore Dynamic and Stochastic IRP with Profits models. It was concluded from the literature that smart routes demand for larger planning periods so the SWCRP should be model as an IRP with Profits, where a larger time horizon will be considered, and the accumulation of weight inside the bins are considered according to the estimates done by the prediction models from T1. The uncertainty inherent to consider days-ahead and the dynamism required in this kind of problem will be tackled in this task. All these features must be modelled, where new mathematical models will be developed.

Task 3-Decision Supporting Tool for the Smart Waste Collection Routing Problem (m 7 to 27). To answer to the [Tool open questions](#), we will design a fast and effective solution method based on meta-heuristics to solve the models developed in T2. Then, from an information management perspective, the requirements from the waste collection companies on which information they need, when and in which format will be gathered through meetings and interviews with waste collection companies’ operations managers, to develop a prototype of a Graphic User Interface.

Task 4-Strategic Location for Sensors Placement (m 13 to 24). To answer to the [Sensor Placement open question](#), we will explore the criteria that influences the decision of sensorizing or not a waste bin and embed those criteria in a decision model to support the decision of which containers should be monitored with static sensors. Having a full sensorized network, where companies typically owned thousands of bins and sensors are still expensive, is very difficult to implement. Moreover, the value of the information transmitted by the sensors will be higher in some waste bins (e.g., more remote bins, with high variability) and lower in others (e.g., non-remote bins with a stable filling rate). Then, not all waste bins in a network are worth to be monitored with a static sensor.

Task 5 - Cost-Benefit Analysis (m 22 to 30). To answer to the [Benefits open question](#), a cost-benefit analysis will be conducted to evaluate the potential market value of this research project. For that, we will test the smart route planning tool with real data (from ERSUC and Valorsul) and assess the benefits of using a system like this when comparing it to the current situation, i.e., blind collection with pre-defined collection frequencies. Historical data from the waste collection companies analysed at T1 will now be used to generate a simulation environment to test four scenarios.

Task 6-Prototype Tool Validation (m 28 to 36). Finally, and to validate the tool developed in a real environment, a testbed will be conducted at ERSUC. Static sensors are going to be installed at a certain number of waste bins of a pilot area (accordingly with the methods developed at T4), and the remaining bins are monitored through mobile sensors or visual observations (depending on the results from T5). Daily collection routes will be defined with the help of our tool.

Fig.1 depicts the complete project framework.

Referências bibliográficas

[1] Ferrer, J. & Alba, E., (2019). BIN-CT: Urban waste collection based on predicting the container fill level. *BioSystems*, 186, 103962

[2] Abbasi, M. & El Hanandeh, A., (2016). Forecasting municipal solid waste generation using artificial intelligence modelling approaches. *Waste Management*, 56, 13–22.

[3] Hannan, M.A., Mamun, M.A. Al, Hussain, A., Basri, H. & Begum, R.A., (2015). A review on technologies and their usage in solid waste monitoring and management systems: Issues and challenges. *Waste Management*, 43, 509–523.

[4] Kannangara, M., Dua, R., Ahmadi, L. & Bensebaa, F., (2018). Modeling and prediction of regional municipal solid waste generation and diversion in Canada using machine learning approaches. *Waste Management*, 74, 3–15.

[5] Rovetta, A., Xiumin, F., Vicentini, F., Minghua, Z., Giusti, A. & Qichang, H., (2009). Early detection and evaluation of waste through sensorized containers for a collection monitoring application. *Waste Management*, 29, 2939–2949.

[6] Gutierrez, J.M., Jensen, M., Henius, M. & Riaz, T., (2015). Smart Waste Collection System Based on Location Intelligence. *Procedia Computer Science*, 61, 120–127

[7] Medvedev, A., Fedchenkov, P., Zaslavsky, A., Anagnostopoulos, T., Khoruzhnikov, S., (2015). Waste Management as an IoT-Enabled Service in Smart Cities, in: Internet of Things, Smart Spaces, and Next Generation Networks and Systems. Springer International Publishing, Cham, pp. 104–115.

[8] Lu, X., Pu, X. & Han, X. (2020). Sustainable smart waste classification and collection system: A bi-objective modeling and optimization approach. *Journal of Cleaner Production*, 276, 124183.

[9] Akbarpour, N., Salehi-Amiri, A., Hajiaghaei-Keshteli, M. & Oliva, D. (2021). An innovative waste management system in a smart city under stochastic optimization using vehicle routing problem. *Soft Computing*, 25, 6707-6727

[10] Saeidi, A., Aghamohamadi-Bosjin, S. & Rabbani, M. (2021). An integrated model for management of hazardous waste in a smart city with a sustainable approach. *Environment, Development and Sustainability*, 23, 10093-10118.

[11] Asimakopoulos, G., Christodoulou, S., Gizas, A., Triantafyllou, V., Tzimas, G., Viennas, E., Gialelis, J., Karadimas, D. & Papalambrou, A., (2015). Towards a Dynamic Waste Collection Management System using Real-time and Forecasted Data, in: Proceedings of the 16th International Conference on Engineering Applications of Neural Networks (INNS). ACM, Rhodes Island, pp. 1–9

[12] Andersson, H., Hoff, A., Christiansen, M., Hasle, G. and Lokketangen, A. (2010). Industrial aspects and literature survey: Combined inventory management and routing, *Computers and Operations Research*, 37, 1515-1536

[13] Baita, F., Ukovich, W., Pesenti, R. & Favaretto, D. (1998). Dynamic Routing-and-Inventory Problems: a review, *Transportation Research Part A: Police and Practice*, 32, 585-598.

[14] Coelho, L. C., Courdeau, J. F. and Laporte, G. (2014). “Heuristics for dynamic and stochastic inventory-routing”, Computers & Operations Research, Vol. 52, pp. 55-67.

[15] Coelho, L. C., Courdeau, J. F. and Laporte, G. (2014). “Thirty years of Inventory Routing”, Transportation Science, Vol. 48 (1), pp. 1-19.

[16] Elbek, M. and Wøhlk, S. (2016). “A variable neighbourhood search for the multi-period collection of recyclable materials”, European Journal of Operational Research, Vol. 249 (2), pp. 540-550.

[17] Mes, M., Schutten, M. and Rivera, A. P. (2014). “Inventory routing for dynamic waste collection”, Waste Management, 34, 1564-1576.

[18] Eurostat, 2022. <https://ec.europa.eu/eurostat/web/products-eurostat-news/-/ddn-20220214-1#:~:text=In%202020%2C%20the%20amount%20of,kg%20more%20than%20in%201995>.

[19] Faloutsos, Christos, et al. "Classical and Contemporary Approaches to Big Time Series Forecasting." Proceedings of the 2019 International Conference on Management of Data. 2019.

[20] Yu, Yong, et al. "A review of recurrent neural networks: LSTM cells and network architectures." Neural computation 31.7 (2019): 1235-1270.

[21] Jozefowiez, N., Semet, F., & Talbi, E.-G. (2008). Multi-objective vehicle routing problems. *European Journal of Operational Research*, 189(2), 293–309.

[22] Montoya, A., Guéret, C., Mendoza, J. E. & Villegas, J. G. (2017). The electric vehicle routing problem with non-linear charging function, *Transportation Research Part B: Methodological*, 103, 87-110.

[Ram1] **Ramos, T.**, de Morais, C. S., **Barbosa-Póvoa, A. P.** (2018). The smart waste collection routing problem: alternative operational management approaches. *Expert Systems with Applications*, 103, 146-158

[Ram2] **Ramos, T.**, Gomes, M.I., **Barbosa-Póvoa, A. P.** (2014). Economic and environmental concerns in planning recyclable waste collection systems. *Transportation Research Part E: Logistics and Transportation Review*, 62, 34-54.

[Jor] **Jorge, D., Antunes, A.P., Ramos, T., Barbosa-Póvoa, A.P.** (2022). A hybrid metaheuristic for smart waste collection problems with workload concerns. *Computers & Operations Research*, 137, 105518

[Mor] Morais, C. S., **Ramos, T., Barbosa-Póvoa, A. P.** (2019) Dynamic approaches to solve the Smart Waste Collection Routing Problem. In: Alves, M.J., Almeida, J.P., Oliveira, J.F. and Pinto, A.A. (Eds). *Operational Research. APDIO 2018. Springer Proceedings in Mathematics & Statistics*. Vol. 278, 173-188. Springer

[Cos] Costa D., Soares J., **Santos P.A.** (2020) Rich and Poor: Simulating Social Policies to Build a Fairer Society. In: Verhagen H., Borit M., Bravo G., Wijermans N. (eds) *Advances in Social Simulation. Springer Proceedings in Complexity*. Springer, Cham.

Publicações anteriores

Ramos, T., de Morais, C. S., Barbosa-Póvoa, A. P. (2018). The smart waste collection routing problem: alternative operational management approaches. *Expert Systems with Applications*, 103, 146-158
<https://www.sciencedirect.com/science/article/pii/S0957417418301404>

Jorge, D., Antunes, A.P., Ramos, T., Barbosa-Póvoa, A.P. (2022). A hybrid metaheuristic for smart waste collection problems with workload concerns. *Computers & Operations Research*, 137, 105518
<https://www.sciencedirect.com/science/article/pii/S0305054821002598>

de Morais, C. S., Ramos, T., Barbosa-Póvoa, A. P. (2019) Dynamic approaches to solve the Smart Waste Collection Routing Problem. In: Alves, M.J., Almeida, J.P., Oliveira, J.F. and Pinto, A.A. (Eds). *Operational Research. APDIO 2018. Springer Proceedings in Mathematics & Statistics*. Vol. 278, 173-188. Springer
https://link.springer.com/chapter/10.1007/978-3-030-10731-4_12

Costa D., Soares J., Santos P.A. (2020) Rich and Poor: Simulating Social Policies to Build a Fairer Society. In: Verhagen H., Borit M., Bravo G., Wijermans N. (eds) *Advances in Social Simulation. Springer Proceedings in Complexity*. Springer, Cham.
https://doi.org/10.1007/978-3-030-34127-5_11

Mota. B., Gomes, M.I., Barbosa-Póvoa, A.P. (2018). Sustainable supply chains: An integrated modeling approach under uncertainty, *Omega*, 77, 32-57
<https://www.sciencedirect.com/science/article/pii/S0305048317304401>

Tarefas

1 Task 1 - Data Analytics and Predictive Models

12 meses

De 02.01.2023 a 01.01.2024

Membros

Pessoas / Mês	Alocados
0.6	Tânia Rodrigues Pereira Ramos Associação do Instituto Superior Técnico para a Investigação e o Desenvolvimento
1.7	Hélio Silva EVOX Technologies

Pessoas / Mês	Alocados
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1.8	Manuel Lopes Instituto de Engenharia de Sistemas e Computadores, Investigação e Desenvolvimento em Lisboa
3	Pedro Alexandre Simões dos Santos Instituto de Engenharia de Sistemas e Computadores, Investigação e Desenvolvimento em Lisboa

Descrição

Task 1 aims to evaluate different data acquisition methods and treat waste bin fill-level data to reduce associated uncertainty.

In this set, three sources of data are considered: **1) drivers records** (from the two engaged companies – ERSUC and Valorsul); **2) static sensors readings** (from Valorsul), and **3) mobile sensors readings**. Regarding drivers’ records, both companies have already these data and will make them available for this project. Regarding static sensors readings, the sensors used are volumetric sensors developed by our partner Evox Technologies, that uses ultrasonic technology and encompass a communication unit and a battery. They measure, communicate and detect events related with the waste container's volume. Since 650 static sensors were installed since 2020 at Valorsul, there is also a huge amount of data available. Regarding the new monitoring method – mobile sensors, no available data exist (since it is new and aimed to be studied further in this project). To collect these data, an experiment will be conducted for 6 months, where a set of drivers will be involved and they will use the mobile sensors every time they visit a bin, besides doing the visual observation. For this experiment, the sensors from Evox will be used, but in a mobile version, i.e., without being fixed/installed. We will use the same bins that have already static sensors installed (from Valorsul), and thus we will be able to have different data (static sensor readings, mobile sensor readings and visual observations) for the same waste bins, and for the same period of time (6 months). This will allow to compare the three data sets, using statistical methods, and able to draw some conclusions.

Besides the comparison between the three data sets, we will perform predictions based on each one of them. For that, data analytics techniques will be applied, ranging from simple temporal series modelling [19], to more complex methods that can be used when more information is available [20]. Models based on LSTM neural networks [Cos] can exploit information about each location, e.g. type of neighbourhood, commercial or residential, day of the week, holidays vs. school days, and others. Then, predictive models will be adjusted to estimate future fill levels for each bin. To validate the predictive models, each source set (drivers records, static sensors readings and mobile sensors readings) will be split into a training set and a testing set, allowing the predictive models to be trained with the data from the training set, and applied to give estimates for the testing set. The ultimate goal of this task is to compare how good are the estimates done by each source of data and conclude on the value of information given (e.g., waste bin visits that could be avoided, overflows that could be avoided, etc.), when proper data analytics and predictive methods are applied.

Task 1 will be mainly performed by INESC-ID team, expert in machine learning techniques. EVOX team will also support this task regarding the sensor data set.

The main deliverables of this task are:

Paper1: Comparison of three monitoring approaches: drivers records, static sensors and mobile sensors

Paper2: New predictive models to estimate waste deposition based on drivers records versus sensors readings.

2 Task 2 - Smart Waste Collection Routing Problem Modelling

21 meses

De 02.01.2023 a 01.10.2024

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Membros

Pessoas / Mês	Alocados
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2.7	Diana Rita Ramos Jorge Associação do Instituto Superior Técnico para a Investigação e o Desenvolvimento
5.85	Tânia Rodrigues Pereira Ramos Associação do Instituto Superior Técnico para a Investigação e o Desenvolvimento

Descrição

Task 2 aims to develop Dynamic and Stochastic IRP with Profits models.

The Smart Waste Collection Routing Problem was introduced in the work of [Ram1]. It was modelled and solved as a Vehicle Routing Problem with Profits (VRPP), which implies a planning horizon of just one day. Given the feedback from the industry along the previous project, and also based on the literature [Mor1], it was concluded that the planning period should be larger than one day. To consider wider planning horizons the problem should be modelled as an Inventory Routing Problem (IRP), where the (estimated) inventory levels for each waste bin, for each day of the planning period are taken into account, coupled with the routing decisions. This was confirmed in the work by [Mor1], where results for the IRP and VRPP approaches using

small-size instances were compared and IRP models provided better results. Therefore, IRP models are deepen explored in this project as firstly as deterministic, being later extended to deal with uncertainty:

1. Deterministic IRPs assume that demand information is known by the decision-maker at the beginning of the planning horizon. The deterministic IRP model proposed by [Mor1] will be deepen in this task.
2. Stochastic IRPs (SIRP) explore data characteristics; data analytics (Task 1) will be used to extract such characteristics from historical data. Uncertainty modelling techniques will be then applied to solve the SIRP. In particular, stochastic or robust optimization will be studied.
3. Dynamic IRPs (DIRP) assume that demand is not completely known at the beginning of the planning horizon but that it will be gradually revealed over time. This characteristic is of special interest to this project, since using real-time information from the sensors naturally enables a “dynamic” context. Dynamic stochastic approaches are then explored, and the work developed previously will be extended to deal with the dynamic nature of the problem considering the uncertain context.

In the previous works [Ram1, Mor1], the objective function considered was Profit Maximization (revenues from selling the recyclable waste minus the transportation cost related to the waste collection). Nevertheless, and from the feedback from practitioners, it became clear that the service level to the population must also be a priority (service level can be measured by the number of overflowing bins and/or imposing a maximum time without visiting a waste bin, that impact public health issues and must be accounted for). Therefore, in this task, this objective will be also considered. Different ways of addressing this objective will be explored: 1) multi-objective approaches [21], where two objectives are at stake – maximize profit and minimize overflows ; 2) maintain a single objective approach but including a cost that represents the service level (in other words, include the cost of overflow).

Mixed-Integer Linear Programming (MILP) models will be developed along points 1) to 3) and different types of formulations will be implemented and tested. The MILP models will be implemented with GAMS language and solved through the CPLEX Optimizer. To test the models, test instances from the previous exploratory research project will be used and also new ones will be generated using the data collected at Task 1 to better represent real waste collection problems.

This task will be mainly performed by CEGIST team, given their knowledge in modelling routing problems.

The main deliverables of this task are:

- Paper3: New models for the Smart Waste Collection Routing Problem: Dynamic and Stochastic IRP

3 Task 3 - Decision Supporting Tool for the Smart Waste Collection Routing Problem

21 meses

De 01.07.2023 a 31.03.2025

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3.6	Vitor Goulão EVOX Technologies
4.6	António José Pais Antunes Universidade de Coimbra
18	BI-a - 03 Bolsa de Investigação - Estud. dout. ou lic. e mestres em CNCG Universidade de Coimbra

Descrição

The models developed in Task 2, due to their complexity, are expected to present two main limitations: 1) can only solve small to medium size instances to optimality, requiring long computational times even to address medium problems, 2) the model language and solvers used cannot be easily supported by the industry. Nevertheless, such models are needed to assess the quality of the solutions when approximate algorithms like heuristic and meta-heuristics are explored, which will be performed in this task. With the use of sensors, routes must be dynamic - every day new and different routes are to be generated and carried out by the drivers. A dynamic route scenario demands for short computational times. Moreover, since this research project intends to produce some knowledge to be transfer to practitioners (through the development of a prototype of a decision support tool to plan smart collection routes), large size problems will be at stake, and the prototype must be developed under a programming language that can be used by companies. Therefore, this task has two goals: develop metaheuristics to solve the IRP and develop a Graphic User Interface prototype for the Decision Supporting Tool, where the metaheuristics developed plus the predictive methods from Task 1 are embedded.

Knowing that routing problems are typically very complex to solve, the problems addressed at Task 2 (Deterministic IRP, Stochastic IRP and Dynamic IRP) will now be solved using heuristics and meta-heuristics [Jor], techniques that have proved to provide good solutions in short period of time within traditional routing problems. Apart from the solution’s time enhancement, is important to prove that the solution method is able to find good solutions (optimal or near-optimal solution). To that end, the test instances devised at Task 2 will be use at Task 3. The solutions obtained will be compared to the results from Task 2 in order to assess how close the heuristic solutions are from the optimal solution. In that way, the effectiveness of the solution approach is tested.

Then, the solution methods developed in this task, plus the predictive methods developed at Task 1 (either for the drivers records historical data and sensors readings historical data) will be embedded in a prototype for the Graphic User Interface (GUI) for the Decision Supporting Tool to plan smart collection routes. To build the GUI, the requirements from the waste collection companies must be gathered through meetings and interviews.

UC will work on the development of heuristics and metaheuristics to the Smart Waste Collection Routing Problem and also with CEGIST to test the solution method using the test instances. EVOX will be responsible for gather the information requirements from the waste collection companies through meetings and interviews, and develop the prototype for the Graphic User Interface accordingly with those requirements.

The main deliverables of this task are:

- Paper 4: New solution methods for the Smart Waste Collection Routing Problem
- Graphic User Interface Prototype for the Decision Supporting Tool

4 Task 4 - Strategic Location for Sensors Placement

12 meses

De 01.01.2024 a 31.12.2024



Membros

Pessoas / Mês	Alocados
0.54	Ana Paula Ferreira Dias Barbosa Povia Associação do Instituto Superior Técnico para a Investigação e o Desenvolvimento
6	BI-b - 01 Bolsa de Investigação - Estud. mestrado, mestrado int. ou lic. e mestre em CNCG Associação do Instituto Superior Técnico para a Investigação e o Desenvolvimento
1.5	Tânia Rodrigues Pereira Ramos Associação do Instituto Superior Técnico para a Investigação e o Desenvolvimento
0.45	Hélio Silva EVOX Technologies
12	BI-a - 02 Bolsa de Investigação - Estud. dout. ou lic. e mestres em CNCG Instituto de Engenharia de Sistemas e Computadores, Investigação e Desenvolvimento em Lisboa
1.8	Manuel Lopes Instituto de Engenharia de Sistemas e Computadores, Investigação e Desenvolvimento em Lisboa
3	Pedro Alexandre Simões dos Santos Instituto de Engenharia de Sistemas e Computadores, Investigação e Desenvolvimento em Lisboa

Descrição

Waste collection systems are complex as they involve thousands of containers. Although monitoring technologies, as volumetric sensors, allow to better understand these systems and to gain insights from the data collected, they are still very pricey. It is then necessary to develop financially more sustainable strategies for integrating sensors into the waste collection activities. For that, in this task, it will be analysed in which circumstances makes sense to have a static sensor embedded in a container to measure its fill level. A set of criteria will be identified taking into account the characteristics of the containers and their relevance for the problem (e.g. location, type of population and facilities nearby, level of predictability, etc.). A decision support model will be proposed, and it must be such that it helps minimizing the investment costs associated with the utilization of the volumetric sensor technology (by monitoring only a reduced number of containers) while maintaining the benefits inherent to its use. Thus, a trade-off analysis between the cost and benefits of monitoring all the containers or only few of them will be performed in order to comprehend the feasibility of the proposed decision support model. The model will select from a full set of containers which of those containers must have their fill level measured in real time through the use of volumetric sensors. Two strategies are foreseen for the algorithms to be developed: 1) Two-step approach: Select Containers First, Route Second; 2) One-step approach: Select + Route Integrated.

In the first strategy, criteria like average fill rate, variance, remoteness, etc., of each container will be considered to select containers through sampling techniques. Then, in the second step, routes will be defined following two approaches: 1) using the predictive methods developed at Task 1 for both sets of data (drivers records for the non-monitoring bins and sensors readings for the monitored bins), and apply the models developed at Task 2 and/or the solution methods developed at Task 3 (depending on the size of the instance) to define a smart collection route plan, which is updated whenever needed; 2) for the non-monitored bins, define the optimal collection frequency and define regular routes for each collection frequency through classical VRP models, and for the monitoring bins, the regular routes will be detoured to collect them when needed. In the latter option, the VRP with detours must be studied [22] as a new option for the definition of smart collection routes when only a sample of bins are being sensorized and where the strategy of regular routes for the non-monitored bins is adopted.

In the second strategy, the selection and routing steps will be integrated, since both decisions have impacts on each other. Building a model that minimize the risk of an earlier collection route (leading to low efficiency) plus the risks of a latter collection route (leading to overflows), subject to a limited budget for the sensor installation could be a way of integrating both decisions.

This task will be mainly performed by CEG-IST and INESC-ID teams.

The main deliverables of this task are:

Paper 5: Decision model for containers selection for monitoring purpose: Two-Step approach

Paper 6: Decision model for containers selection for monitoring purpose: One-Step approach

5 Task 5 – Cost-Benefit Analysis

9 meses

De 01.10.2024 a 30.06.2025



Membros

Pessoas / Mês	Alocados
0.9	Ana Paula Ferreira Dias Barbosa Povia Associação do Instituto Superior Técnico para a Investigação e o Desenvolvimento
8.25	BI-a - 01 Bolsa de Investigação - Estud. dout. ou lic. e mestres em CNCG Associação do Instituto Superior Técnico para a Investigação e o Desenvolvimento
2.55	Tânia Rodrigues Pereira Ramos Associação do Instituto Superior Técnico para a Investigação e o Desenvolvimento
6	BI-a - 02 Bolsa de Investigação - Estud. dout. ou lic. e mestres em CNCG Instituto de Engenharia de Sistemas e Computadores, Investigação e Desenvolvimento em Lisboa
0.6	Manuel Lopes Instituto de Engenharia de Sistemas e Computadores, Investigação e Desenvolvimento em Lisboa
0.9	Pedro Alexandre Simões dos Santos Instituto de Engenharia de Sistemas e Computadores, Investigação e Desenvolvimento em Lisboa
1.34	António José Pais Antunes Universidade de Coimbra
5.25	BI-a - 03 Bolsa de Investigação - Estud. dout. ou lic. e mestres em CNCG Universidade de Coimbra

Descrição

Task 5 aims to perform a cost-benefit analysis, where four scenarios will be studied, choosing pilot-areas from our partner companies ERSUC and Valorsul. Simulations with real data regarding the daily fill levels for all containers inside the pilot-area will be used (already analysed in Task 1). Multi-agent simulation techniques similar to the ones used in [Cos20] are foreseen to be used in this task. The four scenarios are the following:

- Scenario 1 represents the current solution performed by the waste collection companies so far, i.e., blind collection without collection frequency optimization neither route optimization. For that, the static routes done made the companies in the pilot-area will be analysed: collection frequency, waste collected, kilometres travelled, fill rates of the containers in the moment of the collection (recorded by the drivers), etc.
- Scenario 2 represents an intermediate solution to assess two effects independently: route optimization and collection frequency optimization, but still in a blind collection context (without any analysis or a priori knowledge of the fill levels). Scenario 2’ represents a blind collection with collection frequency optimization but no route optimization, i.e., the same static routes are performed as in Scenario 1, but with an optimized frequency. Scenario 2’’ represents a blind collection with route optimization but no collection frequency optimization (the same collection frequency as in Scenario 1). To optimize the routes, the models from previous work of [Ram2] will be applied. Scenario 2’’’ represents blind collection with collection frequency and route optimization. This scenario will be the baseline for the benefits assessment of a smart collection versus a blind collection.
- Scenario 3 represents a smart collection, with a full sensorized network combined with route optimization. For that, the IRP models and solution methods developed at Tasks 2 and 3 will be applied for planning the routes (depending on the size of the real instance both methods can be used, or just the second one for large size instances), and the predictive method from Task 1 will be also use to estimate the fill levels for the days ahead as needed for the IRP. In this scenario, data from the sensors will be simulated based on real data, as said before.
- Scenario 4, similar to scenario 3 but with real-time information obtained from a sample of containers. In this case, the decision models developed at Task 4 are applied to decide on which containers should be monitored with static sensors. The two strategies devised for Task 4 will be tested here. For the remaining waste bins, the other two monitoring approaches will be considered: mobile sensors and visual observations.

The benefits will be assessed in two criteria, as mentioned in Task 2 – Profit and Service Level provided to the population (measured by the number of overflows). After assessing the benefits of collection and route optimization over the current situation (scenario 2 vs. scenario 1), the benefits of having access to real-time information of every containers (scenario 3 vs. scenario 2), and the benefits of having access to real-time information of some containers (scenario 4 vs. scenario 3), the costs of having access to real-time information (equipment, batteries, communications, maintenance, etc.) will be computed to evaluate if the benefits are higher than the costs.

This task will be mainly performed by CEGIST team, INESC-ID team and UC team.

The main deliverable of this task is:

Paper 7: Cost-Benefit Analysis: As-Is (Blind Collection) versus To-Be (Smart Collection) in two real case studies

Membros

Pessoas / Mês	Alocados
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Pessoas / Mês	Alocados
0.45	Ana Paula Ferreira Dias Barbosa Pova Associação do Instituto Superior Técnico para a Investigação e o Desenvolvimento
6.75	BI-a - 01 Bolsa de Investigação - Estud. dout. ou lic. e mestres em CNCG Associação do Instituto Superior Técnico para a Investigação e o Desenvolvimento
2.1	Tânia Rodrigues Pereira Ramos Associação do Instituto Superior Técnico para a Investigação e o Desenvolvimento
1.2	Hélio Silva EVOX Technologies
1.8	Vitor Goulão EVOX Technologies
6	BI-a - 02 Bolsa de Investigação - Estud. dout. ou lic. e mestres em CNCG Instituto de Engenharia de Sistemas e Computadores, Investigação e Desenvolvimento em Lisboa
0.6	Manuel Lopes Instituto de Engenharia de Sistemas e Computadores, Investigação e Desenvolvimento em Lisboa
1.02	Pedro Alexandre Simões dos Santos Instituto de Engenharia de Sistemas e Computadores, Investigação e Desenvolvimento em Lisboa
0.9	António José Pais Antunes Universidade de Coimbra
6.75	BI-a - 03 Bolsa de Investigação - Estud. dout. ou lic. e mestres em CNCG Universidade de Coimbra

Descrição

Task 6 is the last research task of the project and integrates all the efforts from the previous tasks. The aim is to test the prototype tool in a real environment. While in Task 5 the developed models were tested in a simulated environment, this task aims to test them in a real environment – at ERSUC. The testbed is planned to be performed between April to December in order to test the system at different times of the year, under different weather conditions and different demand volumes. The same pilot-area will be used, where the models developed at Task 4 are applied to decide the static sensor placement in that pilot-area (if the same pilot-area is used, the result was already obtained at Task 4); then, those sensors will be actually installed (we are estimating to install 75 sensors), and the remaining bins are monitored with mobile sensors or visual observations (depending on the results from the previous task). Then, the prototype of the smart waste collection planning tool will be used to inform the company of the collection routes to perform in each day. A close relationship with our partner ERSUC is going to be established during this Task, in order to validate and evaluate the routes suggested by our planning tool. The prototype tool represents an output of this research project that can be immediately usable, with impact to society. The resulting prototype tool will be made available free of charge to waste management companies in the framework of collaboration protocols. Later it can be extended to other areas (e.g., supply system of Liquefied Petroleum Gas tanks, where sensors can monitoring LPG levels in customer’s reservoirs and the supply routes be improved in terms of efficiency and service level – prevention of stockouts).

This task will be mainly performed by EVOX and UC teams, but all teams will be involved.

The main deliverable of this task is:

- Technical Report: Results from the prototype tool testbed

7 Project Management

36 meses

De 02.01.2023 a 01.01.2026



Membros

Pessoas / Mês	Alocados
0.36	Ana Paula Ferreira Dias Barbosa Pova Associação do Instituto Superior Técnico para a Investigação e o Desenvolvimento
1.8	Tânia Rodrigues Pereira Ramos Associação do Instituto Superior Técnico para a Investigação e o Desenvolvimento
0.3	Hélio Silva EVOX Technologies
1.08	Pedro Alexandre Simões dos Santos Instituto de Engenharia de Sistemas e Computadores, Investigação e Desenvolvimento em Lisboa

Pessoas / Mês	Alocados
0.36	António José Pais Antunes Universidade de Coimbra

Descrição

This task is of great importance in a 3-year research project, with 7 different entities involved (CEG-IST, INESC-ID, UC, EVOX, University of Bologna, ERSUC and Valorsul). The complexity of the project, the high number of stakeholders involved and the uncertainty naturally involved in a research project demands for a well-defined project management plan. This task aims to plan, organize, coordinate, integrate and control all the remaining tasks, and identify, plan and manage the project risks, so that the project is completed successfully.

The principal investigator (PI) Tânia Ramos, who also holds the Project Management Professional (PMP) certification, will be the responsible person of this task, with the help from the co-PI (Pedro Santos), as well as from the task owners.

The project will be monitored on a regular basis to track the overall progress in terms of scope, time, cost, and risk management. Scope management will ensure that all the scientific work is done meeting high-quality standards to accomplish the goals of the project and achieve the expected results in terms of indicators. Time management will ensure a timely completion of the project, where the critical path and milestones achievement will be managed carefully. Cost management will ensure that the project will be completed within the approved budget, where the actual costs will be monitored and compared with the budget plan. Risk management will ensure that the project risks are identified and monitored along with the project, and proper mitigation and contingency plans are devised and applied when needed.

Potential risks anticipated at this stage are the following (only the critical points are listed):

- Risk 1. Difficulty or delay in finding the right candidates for the PhD grants
- Risk 2. The waste company ERSUC drops out, jeopardizing the testbed task
- Risk 3. Delay in getting real data from the waste management companies
- Risk 4. Delay in the sub-tasks assigned to the industrial partner EVOX
- Risk 5. Delay in installing the sensors for the testbed

The mitigation measures and the contingency plan for each risk are the following:

Risk 1. Early identification of potential candidates among students with whom the research team as frequent contact (all teams members from the universities teach at the second and third cycle).

Risk 2. Careful engagement of ERSUC, discussing the intermediate findings (in a high-light fashion) regularly; if the risk identified happens, the other waste company engaged (Valorsul) could serve as a testbed, using a different geographical area than the one that already has sensors installed.

Risk 3. Start working with data from previous projects and simulate data based on that.

Risk 4. Add one full dedicated resource to execute the sub-tasks assigned to the industrial partner (as self-financing from the company).

Risk 5. Start the testbed with a lower number of sensors.

A good communication workflow between all team members will help detecting deviations to the plan in an early stage and corrective actions can be implemented in a fast way. Each team owner will guarantee the sharing of information amongst the researchers involved in their respective task. Every month the task’s leaders will send the PI a synthetic overview of each task development with the main highlights, major opportunities and threats according to the work plan. The PI will compile these overview reports, which will be discussed in the team meetings. The team meetings will happen once a month (virtual meetings) and every trimester a presential meeting will occur, in a rotation scheme: CEG-IST (Lisbon) -> University of Coimbra (Coimbra) -> INESC-ID (Lisbon) -> Evox (Castelo Branco). Thus, 12 presential meetings are foressen.

Calendarização e gestão do projeto

Lista de milestones

Data	Designação	Descrição
31.12.2023	M1 - Prediction Models Developed	At this point, data analysis for the three data sets (static sensors readings, mobile sensors readings and drivers records) is concluded and the prediction models are developed.
30.09.2024	M2 - Smart Routing Models Developed	At this point, the Dynamic and Stochastic Inventory Routing with Profit models for solving the Smart Waste Collection Routing Problem are concluded.
31.12.2024	M3 - Sensor Placement Models Developed	At this point, the methods to select the containers that should be sensorized through static sensors are developed.
31.03.2025	M4 - Decision Support Tool Developed	At this point, a prototype of the decision supporting tool for the dynamic route planning is devised based on the new solution methods developed.
30.06.2025	M5 - Cost-Benefit Analysis Concluded	At this point, a cost-benefit analysis is concluded for both ERSUC and Valorsul case-studies, using a simulation environment.

Data	Designação	Descrição
31.12.2025	M6 - Test-Bed Concluded	This is the final milestone and represents the project completion. At this point, all tasks are concluded, including the prototype validation in a real environment (ERSUC case study).

Cronograma

[timeline Projetos FCT_WSmart Route+.pdf](#)

Obtido em 09-03-2022 pelas 09:52:27

Descrição da estrutura de gestão

PROJECT STEERING COMMITTEE: This committee comprises the principal investigator (PI) Tânia Ramos, the Co-PI Pedro Santos, and the tasks owners: Ana Póvoa (AP), António Pais Antunes (APA), and Hélio Silva (HS). The steering committee will provide executive oversight, advocacy, support and decision making regarding major risks of the project.

METHODS FOR COORDINATING ALL PARTICIPANTS AND REPORTING STRUCTURE: Each task owner will guarantee the sharing of information amongst the researchers involved in their respective task and will report to PI that will guarantee tasks through an efficient flow of information.

Every month the task’s leaders will send the PI a synthetic overview of each task development with the main highlights, major opportunities and threats according to the work plan. The PI will compile these overview reports, which will be discussed in the team meetings.

COMMUNICATION WITH THE TEAM MEMBERS: The team will have internal meetings every month (virtual and presential ones). The international consultant Prof. Daniele Vigo will be invited to participate at some of those meetings. Moreover, a shared folder will be created to share the main documentation of the project. A Skype group or a WhatsApp group will be created to facilitate the communication between team members.

Questões éticas

Existem questões éticas identificadas neste projeto?

Não

Declarações de ética que considera apropriadas

-

Fundamentação

-

Agenda 2030

Enquadramento da candidatura nos OSD da Agenda 2030

- ODS 11: Cidades e Comunidades Sustentáveis
- ODS 9: Indústria, Inovação e Infraestruturas
- ODS 6: Água Potável e Saneamento

Fundamentação

The WSmart Route+ project aligns with SDG 11 Sustainable Cities and Communities, SDG 9 Industry, Innovation and Infrastructure, and SDG 6 Clean Water and Sanitation as it studies a new paradigm on smart waste management. Data analytics combined with route optimization are explored to improve waste collection operation supporting the adoption of a smart collection based on dynamic routes as opposed to the traditional blind collection based on static routes. WSmart Route+ objectives and scientific challenges are: 1) the development of a planning tool combining innovative methods to deal with the complexity related with the definition of smart collection routes, considering data provided by sensors (static and mobile) and by driver’s visual observations; 2) the development of innovative models to decide in which containers static sensors should be installed; 3) the development of a cost-benefit analysis comparing the as-is to the to-be situation in real case studies; 4) the validation of the planning tool prototype with a testbed in a real environment. The ultimate goal of this research is to increase resources efficiency and improve the service level provided to the communities (basic sanitation) in waste management systems towards a more sustainable development.

WSmart Route+ contributes to SDG 9, namely to target 9.4, through the development of tools that will make the waste collection industry more sustainable, increasing resource-use efficiency. By using our tools, companies will avoid deadhead situations (travelling many kilometres with no (or few) cargo) and low vehicle usage situations, optimizing resources usage and reducing CO2 emissions. The CO2 emissions per unit of value-added indicator (9.4.1) will significantly decrease.

Moreover, having more precise information on the amount of waste existing in each waste bin will also contribute to SDG 6, namely, to target 6.b, given that WSmart Route+ will support and strengthen the participation of local communities (municipalities) in improving sanitation management. Waste bins will be visited when justified in terms of hygiene and public health, i.e., waste bins cannot exceed their maximum capacity to avoid the accumulation of waste around the containers. Thus, waste collection routes will be planned considering more "intelligent" criteria, also preventing containers from being collected when they are already overflowing, with the harmful consequences to public health.

WSmart Route+ will contribute to SDG 11, namely target 11.6, as it pays special attention to municipal waste and its adverse environmental impacts on cities. The urban solid waste regularly collected (indicator 11.6.1) is expected to improve, given that our planning tool will indicate the “best” collection frequency (when to visit waste bins) taking into account both resource efficiency issues and public health issues.

Outros projetos financiados

MIT-EXPL/SUS/0132/2017		FCT	
IR		Tânia Rodrigues Pereira Ramos	
Estado do projeto		Homologado	
Título do projeto		(em inglês)	
WSMART ROUTE – Waste Collection based on a Real Time Route Planning System			
Instituição proponente		Associação do Instituto Superior Técnico para a Investigação e o Desenvolvimento	
Financiamento total		70 581,00 €	
Data de início		Duração	
03.09.2018		(meses)	
		20	
Principais resultados do projeto que considera relevantes para esta candidatura			
The main results relevant for this application are:			
- Establish a minimum fill level limit to trigger a collection is not a good strategy; even containers with low filling levels can “be worth it” to be collected [Ram18] - The collection route to be performed today should consider an estimate of the future fill levels [Mor19] - In the real case studied (for one municipality, and one recyclable material), using smart collection strategies can reduce the number of routes performed in 33%, and reduce the distance travelled in 30% [Mor22]			
These represent the basis/foundation of the WSmart Route+ project.			

PTDC/CCI-COM/30787/2017		FCT	
IR		Pedro Alexandre Simões dos Santos	
Estado do projeto		Homologado	
Título do projeto		(em inglês)	
Social Learning for Intelligent Characters			
Instituição proponente		Universidade de Lisboa (UL) - Instituto Superior Técnico - Instituto de Engenharia de Sistemas e Computadores - Instituto de Engenharia de Sistemas e Computadores, Investigação e Desenvolvimento em Lisboa	
Financiamento total		239 533,78 €	
Data de início		Duração	
01.10.2018		(meses)	
		48	
Principais resultados do projeto que considera relevantes para esta candidatura			
In the SLICE project we developed state-of-the-art neuro-symbolic techniques to analyze dialogue or narrative text and generate an answer. Some of these techniques are based on predictive models and/or LSTMs [Cos20] and as such will also be used in the development of the predictive models in Task 1 of the WSmart Route+ project.			

Ficheiros anexos

[Endorsement WSmart Route+Research Project_Valorsul.pdf](#)

Obtido em 09-03-2022 pelas 13:12:54

[Supporting Letter_ERSUC.pdf](#)

Obtido em 09-03-2022 pelas 13:13:00

[Letter of Collaboration_DanieleVigo.pdf](#)

Obtido em 10-03-2022 pelas 11:57:12

[Figure 1.pdf](#)

Obtido em 10-03-2022 pelas 11:58:58



Indicadores de realização previstos

Descrição	2023	2024	2025	2026	Total
A • Publicações	0	0	0	0	
Livros	0	0	0	0	0
Capítulos de livros	0	0	0	0	0
Artigos em revistas internacionais	0	2	5	0	7
Artigos em revistas nacionais	0	0	0	0	0
B • Comunicações	0	0	0	0	
Comunicações em encontros nacionais	0	1	1	0	2
Comunicações em encontros científicos internacionais	0	3	5	0	8
C • Relatórios	1	1	2	0	4
D • Organização de seminários e conferências	0	0	1	0	1
E • Formação avançada	0	0	0	0	
Teses de Doutoramento	0	0	3	0	3
Teses de mestrado	0	2	2	0	4
Outras	0	0	0	0	0
F • Modelos	0	2	0	0	2
G • Aplicações computacionais	0	0	1	0	1
H • Instalações piloto	0	0	1	0	1
I • Protótipos laboratoriais	0	0	0	0	0
J • Patentes	0	0	0	0	0
K • Outros	0	0	0	0	
	0	0	0	0	0
	0	0	0	0	0
	0	0	0	0	0

Divulgação

Ações de divulgação da atividade científica previstas no projeto

The dissemination plan is as follows:

- Scientific Papers and Oral presentations in National and International Conferences

Throughout the project, some time will be dedicated to publishing the research results in international research journals and present them on international conferences. Seven research publications in highly ranked scientific journals and ten oral presentations in international conferences are expected to result from this project within the planned horizon. Nevertheless, the excellence of the research team inspires confidence that the number of publications related to this project will grow in the following years. The preprint manuscripts will be uploaded at the RCAAP.

- Webpage, Facebook account and a Twitter account

A project’s Webpage will be created, and the Facebook and Twitter accounts of IST and CEGIST will be used to share the news and the results of the project with the scientific community, industry practitioners and other stakeholders from the solid waste sector and the IT field.

- Workshop

The final output of this project is expected to be of great value to industry practitioners as well as to the scientific community. A one-day workshop, which will take place at the end of the project in Lisbon, will gather team members and industrial experts from the municipal solid waste sector. This will be a space for sharing the main results achieved throughout the project and set the basis for the product development.

- Participation in Outreach events

Participation in several public events to share the results of the project with general public and civil society are also foreseen. Some events were already identified:

“90 seconds of Science” | The radio program that explains science made in Portugal with the objective of improving science communication and contributing to the increase of scientific literacy of the general public. The episodes feature Portuguese scientists talking about their research work and projects conducted in universities and research centers in Portugal and abroad.

“European Researchers’ Night” | The European Researchers' Night is a Europe-wide public event, which displays the diversity of science and its impact on citizens' daily lives in fun, inspiring ways.

FIC.A - Festival Internacional de Ciência de Oeiras | Festival of science and technology developed in Portugal, with a strong focus on education, culture and art.

Instituição proponente

Associação do Instituto Superior Técnico para a Investigação e o Desenvolvimento

Rubrica	2023	2024	2025	2026	Total
Recursos Humanos	16 136,00 €	22 591,88 €	16 136,00 €	0,00 €	54 863,88 €
Missões	1 200,00 €	2 700,00 €	2 700,00 €	0,00 €	6 600,00 €
Justificação do financiamento solicitado Missions include: - Consultant: The project has a consultant (Prof.Daniele Vigo, from University of Bologna). It is planned to have his presence once a year in our project meetings. Therefore, 3 trips (Bologna-Lisbon) will be supported by the project (750€ x 3 trips = 2250€) - Internal meetings: The research team of this project gathers researchers from 3 different cities (Lisbon, Coimbra and Castelo Branco). Although modern technologies will be used intensively for virtual meetings, personal meetings are mandatory. Twelve meetings are planned: six in Lisbon, three in Coimbra and three in Castelo Branco. Therefore, CEGIST will have 6 travels : 3 participants x 50€ (train ticket + meal allowance) x 6= 900€ - Fieldwork: Task 1, Task 5 and Task 6 are extremely demanding for fieldwork for data collecting, observation, sensors installation and meeting with our partners ERSUC(located at Coimbra) and Valorsul (located at Cadaval). We expect 3 fieldwork travels for CEGIST with a cost of 150€ each (3 x 150€=450€) - International missions: This funding will be used for travel, accommodation and registration fees for the researchers in the project to attend top international conferences. For CEGIST, one international conference in 2024 and one international in 2025 are budgeted (2 international meetings x 1500€ = 3000€).					
Instrumentos e equipamento científico e técnico	1 500,00 €	0,00 €	0,00 €	0,00 €	1 500,00 €
Justificação do financiamento solicitado One laptop computer is required as necessary to support the work development.This will be assigned to CEGIST PhD student.					
Subcontratos	0,00 €	0,00 €	0,00 €	0,00 €	0,00 €
Registo de patentes	0,00 €	0,00 €	0,00 €	0,00 €	0,00 €
Demonstração, Promoção e Divulgação	500,00 €	2 250,00 €	4 250,00 €	0,00 €	7 000,00 €
Justificação do financiamento solicitado Demonstration, Promotion and Dissemination include: - Logo and webpage: The project will have a logo and a webpage for outreach purpose. It was budgeted 500€ in the first year of the project - Workshop: As established in the dissemination strategy, a final workshop in Lisbon is planned for the final results dissemination. A budget of 2000€ in the last year of the project was considered to support the renting of facilities, coffee-breaks and other organizational expenses - Conferences: Presenting the results of the project at top international and national conferences is essential to validate and disseminate the work done. We are expecting to do 10 oral presentations in 6 conferences: Euro, Verolog, APDIO, EurOMA, AAMAS and Wastes (some conferences are held annually, meaning that the participation will happen in 2024 and 2025). Each conference in Europe was assigned a budget of 1500€ (includes travel, accommodation and registration fee) and for national conferences 750€. For CEGIST we will have: 2 national conferences x 1 participant x 750€ = 1500€, plus 2 international conferences x 1 participant x 1500€ = 3000€					
Adaptação de edifícios e instalações	0,00 €	0,00 €	0,00 €	0,00 €	0,00 €
Aquisição de bens e serviços	0,00 €	200,00 €	200,00 €	400,00 €	800,00 €
Justificação do financiamento solicitado Purchase of Goods and Services include English editing for the scientific papers, software maintenance licenses, books and other supporting material for each research center. For CEGIST it was budgeted 800€.					
Gastos gerais	4 834,00 €	6 935,47 €	5 821,50 €	100,00 €	17 690,97 €
Total	24 170,00 €	34 677,35 €	29 107,50 €	500,00 €	88 454,85 €

Instituições participantes

Instituto de Engenharia de Sistemas e Computadores, Investigação e Desenvolvimento em Lisboa

Rubrica	2023	2024	2025	2026	Total
Recursos Humanos	16 136,00 €	16 136,00 €	16 136,00 €	0,00 €	48 408,00 €
Missões	350,00 €	1 850,00 €	1 850,00 €	0,00 €	4 050,00 €
Justificação do financiamento solicitado Missions include: - Internal meetings: The research team of this project gathers researchers from 3 different cities (Lisbon, Coimbra and Castelo Branco). Although modern technologies will be used intensively for virtual meetings, personal meetings are mandatory. Twelve meetings are planned: six in Lisbon, three in Coimbra and three in Castelo Branco. Therefore, INESC-ID will have 6 travels : 2 participants x 50€ (train ticket + meal allowance) x 6= 600€ - Fieldwork: Task 1, Task 5 and Task 6 are extremely demanding for fieldwork for data collecting, observation, sensors installation and meeting with our partners ERSUC(located at Coimbra) and Valorsul (located at Cadaval). We expect 3 fieldwork travels for INESC-ID with a cost of 150€ each (3 x 150€=450€) - International missions: This funding will be used for travel, accommodation and registration fees for the researchers in the project to attend top international conferences. For INESC-ID, one international conference in 2024 and one international in 2025 are budgeted (2 international meetings x 1500€ = 3000€).					
Instrumentos e equipamento científico e técnico	1 500,00 €	0,00 €	0,00 €	0,00 €	1 500,00 €
Justificação do financiamento solicitado One laptop computer is required as necessary to support the work development.This will be assigned to INESC-ID PhD student.					
Subcontratos	0,00 €	0,00 €	0,00 €	0,00 €	0,00 €
Registo de patentes	0,00 €	0,00 €	0,00 €	0,00 €	0,00 €
Demonstração, Promoção e Divulgação	0,00 €	1 500,00 €	1 500,00 €	0,00 €	3 000,00 €
Justificação do financiamento solicitado Conferences: Presenting the results of the project at top international and national conferences is essential to validate and disseminate the work done. We are expecting to do 10 oral presentations in 6 conferences: Euro, Verolog, APDIO, EurOMA, AAMAS and Wastes (some conferences are held annually, meaning that the participation will happen in 2024 and 2025). Each conference in Europe was assigned a budget of 1500€ (includes travel, accommodation and registration fee) and for national conferences 750€. For INESC-ID we will have 2 international conferences x 1 participant x 1500€ = 3000€					
Adaptação de edifícios e instalações	0,00 €	0,00 €	0,00 €	0,00 €	0,00 €
Aquisição de bens e serviços	200,00 €	200,00 €	200,00 €	0,00 €	600,00 €
Justificação do financiamento solicitado Purchase of Goods and Services include English editing for the scientific papers, software maintenance licenses, books and other supporting material for each research center. For INESC-ID it was budgeted 600€.					
Gastos gerais	4 546,50 €	4 921,50 €	4 921,50 €	0,00 €	14 389,50 €
Total	22 732,50 €	24 607,50 €	24 607,50 €	0,00 €	71 947,50 €

Universidade de Coimbra

Rubrica	2023	2024	2025	2026	Total
Recursos Humanos	7 586,00 €	15 172,00 €	15 307,00 €	0,00 €	38 065,00 €
Missões	450,00 €	1 950,00 €	1 950,00 €	0,00 €	4 350,00 €

Rubrica	2023	2024	2025	2026	Total
Justificação do financiamento solicitado Missions include: - Internal meetings: The research team of this project gathers researchers from 3 different cities (Lisbon, Coimbra and Castelo Branco). Although modern technologies will be used intensively for virtual meetings, personal meetings are mandatory. Twelve meetings are planned: six in Lisbon, three in Coimbra and three in Castelo Branco. Therefore, U.Coimbra will have 9 travels : 2 participants x 50€ (train ticket + meal allowance) x 9= 900€ - Fieldwork: Task 1, Task 5 and Task 6 are extremely demanding for fieldwork for data collecting, observation, sensors installation and meeting with our partners ERSUC(located at Coimbra) and Valorsul (located at Cadaval). We expect 3 fieldwork travels for U.Coimbra with a cost of 150€ each (3 x 150€=450€) - International missions: This funding will be used for travel, accommodation and registration fees for the researchers in the project to attend top international conferences. For U.Coimbra, one international conference in 2024 and one international in 2025 are budgeted (2 international meetings x 1500€ = 3000€).					
Instrumentos e equipamento científico e técnico	1 500,00 €	0,00 €	0,00 €	0,00 €	1 500,00 €
Justificação do financiamento solicitado One laptop computer is required as necessary to support the work development.This will be assigned to U.Coimbra PhD student.					
Subcontratos	0,00 €	0,00 €	0,00 €	0,00 €	0,00 €
Registo de patentes	0,00 €	0,00 €	0,00 €	0,00 €	0,00 €
Demonstração, Promoção e Divulgação	0,00 €	2 250,00 €	2 250,00 €	0,00 €	4 500,00 €
Justificação do financiamento solicitado Conferences: Presenting the results of the project at top international and national conferences is essential to validate and disseminate the work done. We are expecting to do 10 oral presentations in 6 conferences: Euro, Verolog, APDIO, EurOMA, AAMAS and Wastes (some conferences are held annually, meaning that the participation will happen in 2024 and 2025). Each conference in Europe was assigned a budget of 1500€ (includes travel, accommodation and registration fee) and for national conferences 750€. For U.Coimbra we will have: 2 national conferences x 1 participant x 750€ = 1500€, plus 2 international conferences x 1 participant x 1500€ = 3000€					
Adaptação de edifícios e instalações	0,00 €	0,00 €	0,00 €	0,00 €	0,00 €
Aquisição de bens e serviços	250,00 €	250,00 €	250,00 €	0,00 €	750,00 €
Justificação do financiamento solicitado Purchase of Goods and Services include English editing for the scientific papers, software maintenance licenses, books and other supporting material for each research center. For U.Coimbra it was budgeted 750€.					
Gastos gerais	2 446,50 €	4 905,50 €	4 939,25 €	0,00 €	12 291,25 €
Total	12 232,50 €	24 527,50 €	24 696,25 €	0,00 €	61 456,25 €

EVOX Technologies

Rubrica	2023	2024	2025	2026	Total
Recursos Humanos	1 740,38 €	3 173,63 €	6 130,69 €	0,00 €	11 044,70 €
Missões	325,00 €	325,00 €	550,00 €	0,00 €	1 200,00 €
Justificação do financiamento solicitado - Internal meetings: The research team of this project gathers researchers from 3 different cities (Lisbon, Coimbra and Castelo Branco). Although modern technologies will be used intensively for virtual meetings, personal meetings are mandatory. Twelve meetings are planned: six in Lisbon, three in Coimbra and three in Castelo Branco. Therefore, EVOX will have 9 travels : 2 participants x 50€ x 9= 900€ x 50% = 450€ - Fieldwork: Task 1, Task 5 and Task 6 are extremely demanding for fieldwork for data collecting, observation, sensors installation and meeting with our partners ERSUC(located at Coimbra) and Valorsul (located at Cadaval). We expect 10 fieldwork travels for EVOX with a cost of 150€ each (10 x 150€=1500€ x 50% = 750€)					
Instrumentos e equipamento científico e técnico	0,00 €	0,00 €	0,00 €	0,00 €	0,00 €

Rubrica	2023	2024	2025	2026	Total
Subcontratos	0,00 €	0,00 €	0,00 €	0,00 €	0,00 €
Registo de patentes	0,00 €	0,00 €	0,00 €	0,00 €	0,00 €
Demonstração, Promoção e Divulgação	0,00 €	0,00 €	0,00 €	0,00 €	0,00 €
Adaptação de edifícios e instalações	0,00 €	0,00 €	0,00 €	0,00 €	0,00 €
Aquisição de bens e serviços	0,00 €	0,00 €	10 150,00 €	0,00 €	10 150,00 €
Justificação do financiamento solicitado The Prototype Tool Validation (Task 6) requires the production of 70 sensors in order to perform a test-bed monitoring 70 containers. It is estimated a cost of 290€ for each sensor that includes the acquisition of all components, assemble and communication costs. A total of 20300€ will be needed, but since EVOX company is only funded at 50%, the cost is 20300€ x 50%=10150€.					
Gastos gerais	516,34 €	874,66 €	4 207,67 €	0,00 €	5 598,67 €
Total	2 581,72 €	4 373,29 €	21 038,36 €	0,00 €	27 993,37 €

EVOX Technologies

AUTO FINANCIAMENTO

Rubrica	2023	2024	2025	2026	Total
Recursos Humanos	1 740,38 €	3 173,63 €	6 130,69 €	0,00 €	11 044,70 €
Missões	325,00 €	325,00 €	550,00 €	0,00 €	1 200,00 €
Instrumentos e equipamento científico e técnico	0,00 €	0,00 €	0,00 €	0,00 €	0,00 €
Subcontratos	0,00 €	0,00 €	0,00 €	0,00 €	0,00 €
Registo de patentes	0,00 €	0,00 €	0,00 €	0,00 €	0,00 €
Demonstração, Promoção e Divulgação	0,00 €	0,00 €	0,00 €	0,00 €	0,00 €
Adaptação de edifícios e instalações	0,00 €	0,00 €	0,00 €	0,00 €	0,00 €
Aquisição de bens e serviços	0,00 €	0,00 €	10 150,00 €	0,00 €	10 150,00 €
Gastos gerais	516,34 €	874,66 €	4 207,67 €	0,00 €	5 598,67 €
Total	2 581,72 €	4 373,29 €	21 038,36 €	0,00 €	27 993,37 €

Orçamento global

Rubrica	2023	2024	2025	2026	Total
Recursos Humanos	41 598,38 €	57 073,51 €	53 709,69 €	0,00 €	152 381,58 €
Missões	2 325,00 €	6 825,00 €	7 050,00 €	0,00 €	16 200,00 €
Instrumentos e equipamento científico e técnico	4 500,00 €	0,00 €	0,00 €	0,00 €	4 500,00 €
Subcontratos	0,00 €	0,00 €	0,00 €	0,00 €	0,00 €
Registo de patentes	0,00 €	0,00 €	0,00 €	0,00 €	0,00 €
Demonstração, Promoção e Divulgação	500,00 €	6 000,00 €	8 000,00 €	0,00 €	14 500,00 €
Adaptação de edifícios e instalações	0,00 €	0,00 €	0,00 €	0,00 €	0,00 €
Aquisição de bens e serviços	450,00 €	650,00 €	10 800,00 €	400,00 €	12 300,00 €
Gastos gerais	12 343,34 €	17 637,13 €	19 889,92 €	100,00 €	49 970,39 €
Total	61 716,72 €	88 185,64 €	99 449,61 €	500,00 €	249 851,97 €

Plano de financiamento

Rubrica	2023	2024	2025	2026	Total
Financiamento FCT	61 716,72 €	88 185,64 €	99 449,61 €	500,00 €	249 851,97 €
Autofinanciamento	2 581,72 €	4 373,29 €	21 038,36 €	0,00 €	27 993,37 €
Outros financiamentos públicos	0,00 €	0,00 €	0,00 €	0,00 €	0,00 €
Outros fundos privados	0,00 €	0,00 €	0,00 €	0,00 €	0,00 €
Total	64 298,44 €	92 558,93 €	120 487,97 €	500,00 €	277 845,34 €



Investigador responsável

Eu, investigador responsável da presente candidatura, declaro:

1- Ter conhecimento de todos os dados e informações constantes da presente candidatura e que me responsabilizo pelo seu conteúdo e veracidade.

2- Ter conhecimento das disposições expressas no Regulamento nº 999/2016, de 31 de outubro, que estabelece as condições de acesso e as regras de apoio a projetos financiados exclusivamente por fundos nacionais através da Fundação para a Ciência e a Tecnologia, I.P. e do Aviso de Apresentação de Candidaturas do Concurso.

3- Não me encontrar em situação de incumprimento injustificado dos requisitos regulamentares no que respeita à apresentação de relatórios de execução científica de projetos concluídos em que também tenha desempenhado o papel de Investigador Responsável.

4- Comprometer-me a assumir a liderança do projeto e responsabilidades decorrentes da sua boa execução.

5- Assegurar o acesso aberto a publicações científicas (peer-reviewed) resultantes da investigação financiada no âmbito deste Aviso para a Apresentação de Candidaturas, de acordo com as Políticas de Acesso Aberto da Fundação para a Ciência e a Tecnologia, I.P..

6- Caso tenha identificado em candidatura ou venham a ser futuramente identificadas questões de ética relacionadas com a execução do projeto, comprometo-me a:

- a) diligenciar junto das instituições beneficiárias no projeto que, durante o período de execução do projeto, terão as permissões necessárias para executar projeto;
- b) recolher toda a documentação necessária e a disponibilizá-la à FCT;
- c) cumprir e a fazer cumprir pela equipa de investigação do projeto, toda a legislação nacional e comunitária aplicável neste domínio a este projeto de investigação.

☒ Tomei conhecimento e concordo com os termos enunciados na declaração de compromisso

Instituição Proponente

Aceite por **Maria Dias** em 21.03.2022 - 16:00 em nome de **Associação do Instituto Superior Técnico para a Investigação e o Desenvolvimento** por delegação competências dos responsáveis da instituição.